

Task 1 Report: Data Gathering and Exploratory Data Analysis

1. Introduction

The objective of this task is to gather relevant customer data, perform exploratory data analysis (EDA), and prepare the dataset for building a predictive model to identify customers at risk of churn. This stage is important because understanding customer behaviour and ensuring high data quality are critical for building reliable machine learning models. The analysis focuses on identifying patterns in customer demographics, transaction behaviour, online activity, and customer service interactions that may indicate potential churn.

2. Data Sources Selected and Rationale

The analysis used multiple datasets from the provided Excel file. Each dataset contributes information about different aspects of customer behaviour.

Customer_Demographics

This dataset contains information about customer characteristics including:

- CustomerID
- Age
- Gender
- MaritalStatus
- IncomeLevel

Rationale:

Demographic information helps identify patterns among different customer groups. For example, age groups, marital status, or income levels may influence customer engagement and the likelihood of churn.

Transaction_History

This dataset records customer purchase activity and includes:

- CustomerID
- TransactionID
- TransactionDate

- AmountSpent
- ProductCategory

Rationale:

Transaction behaviour provides insight into how frequently customers interact with services and how much they spend. Customers with declining spending or fewer transactions may be at higher risk of leaving.

Customer_Service

This dataset captures interactions between customers and support services.

Rationale:

Frequent service interactions or unresolved issues may indicate customer dissatisfaction, which can increase churn risk.

Online_Activity

This dataset tracks customer engagement with digital services such as login frequency and service usage.

Rationale:

Digital engagement is an important indicator of customer involvement. Customers who rarely log in or use services may be less engaged and more likely to churn.

Churn_Status

This dataset contains the target variable indicating whether a customer has churned.

Rationale:

This variable is used as the prediction target for future machine learning models.

3. Data Integration

The datasets were combined using **CustomerID**, which acts as a unique identifier across all tables.

Since the transaction and customer service datasets contain multiple records per customer, these datasets were aggregated to create customer-level features such as:

- Total number of transactions
- Total amount spent
- Number of customer service interactions

After aggregation, all datasets were merged into a single master dataset where each row represents one customer.

This integration allows customer demographics, behaviour, and service interaction data to be analysed together.

4. Exploratory Data Analysis

EDA was performed to understand the structure of the dataset, identify patterns, and detect anomalies.

Descriptive Statistics

Descriptive statistics were used to summarise key numerical variables such as age, transaction amounts, and service interactions. Metrics such as mean, median, minimum, maximum, and standard deviation were calculated to understand the central tendency and spread of the data.

Histogram Analysis

Histograms were used to analyse the distribution of numerical variables such as **Age, Login Frequency, and Total Amount Spent**. These visualisations helped identify how values are distributed across customers and whether the data is skewed or evenly spread.

For example, the histogram of **customer age** shows the distribution of customers across different age groups. Understanding the age distribution helps determine whether certain age groups are more likely to churn.

Similarly, the histogram for **transaction spending** highlights the range of spending behaviour among customers and reveals whether most customers spend small or large amounts. Identifying these patterns can help determine whether lower spending activity is associated with churn.

Histograms also helped detect any unusual values or extreme observations that may require further investigation during data cleaning.

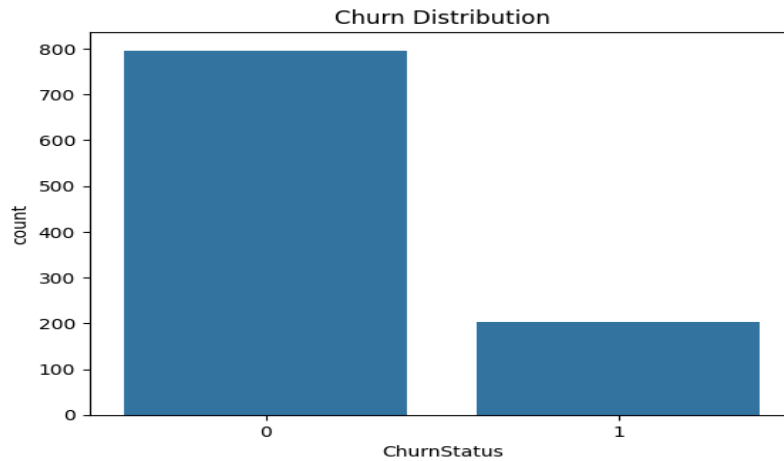


Fig 1: Churn Status v/s Count

Box Plot Analysis

Box plots were used to examine the spread of numerical variables and identify potential outliers. These visualisations display the median, quartiles, and extreme values of the data.

For example, box plots comparing **total spending against churn status** allow us to observe whether customers who churn tend to have different spending patterns compared to customers who remain active.

Box plots also help detect **outliers**, which are unusually high or low values that may influence statistical analysis or machine learning models. Identifying these outliers allows analysts to decide whether to remove, cap, or transform extreme values during the data preprocessing stage.

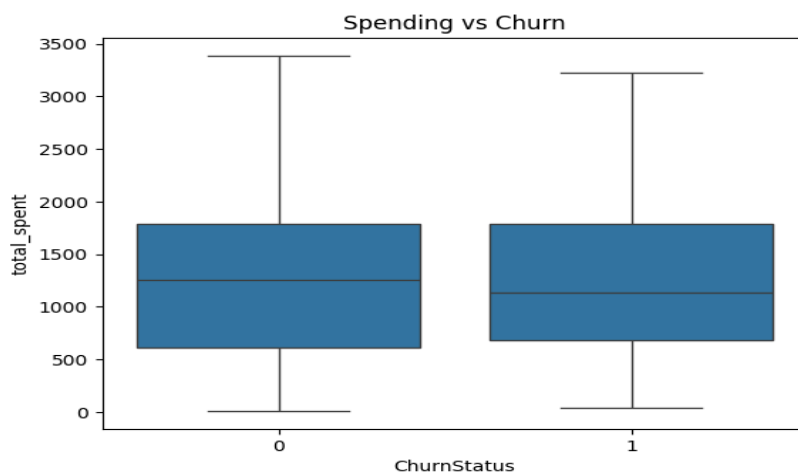


Fig 2:Churn Status v/s Total Spent

5. Insights from Visual Analysis

The visual analysis provided several insights into customer behaviour:

- Spending patterns vary widely among customers.
- Some customers display unusually high or low spending behaviour, indicating potential outliers.
- Differences in behaviour between churned and non-churned customers can be observed when comparing distributions using box plots.

These insights helped guide the **data cleaning and preprocessing steps**, ensuring the dataset is suitable for machine learning modelling.

6. Data Cleaning and Preprocessing

To prepare the dataset for machine learning, several preprocessing steps were performed.

Handling Missing Values

Missing values were addressed using appropriate strategies:

- Numerical values were filled using statistical measures such as the median.
- Some behavioural variables that represented counts were filled with zero where appropriate.

This ensured the dataset remained complete without introducing bias.

Outlier Detection

Outliers were detected using box plots and statistical methods such as the interquartile range (IQR). Extreme values were carefully reviewed and capped where necessary to reduce their influence on model training.

Encoding Categorical Variables

Categorical variables such as gender, marital status, and income level were converted into numerical form using **one-hot encoding**. This step ensures compatibility with machine learning algorithms.

Feature Scaling

Numerical features were standardised to ensure that variables with larger numerical ranges do not dominate the model. Standardisation transforms the data to have a mean of zero and a standard deviation of one.

7. Final Dataset Preparation

After completing the cleaning and preprocessing steps:

- Missing values were handled
- Outliers were treated
- Categorical variables were encoded
- Numerical features were standardised

The final cleaned dataset was exported and is now ready for the next stage of the project: building a machine learning model to predict customer churn.

8. Conclusion

This task established a strong foundation for developing a customer churn prediction model. By carefully selecting relevant data sources, performing exploratory data analysis, and preparing a high-quality dataset, valuable insights into customer behaviour were obtained.

The cleaned dataset will support the development of predictive models that help SmartBank identify customers at risk of leaving and implement targeted retention strategies.